

Proofs

Theorem 1.1. (The Division Theorem for $\mathbb{Z}$): $a, b \in \mathbb{Z}$ with $b > 0$ $\exists! q, r \in \mathbb{Z}$ with $0 \leq r < b$ s.t. $a = bq + r$.

Proof. (Existence): Let $R = \{a - bm : m \in \mathbb{Z} \text{ and } a - bm \geq 0\}$, then as: if $a \geq 0$, $a - b \cdot 0 = a \geq 0$ so $a \in R$; if $a < 0$, $a - b \cdot a = a(1 - b) \geq 0$ so $a - b \cdot a \in R$, $R \neq \emptyset$. Thus, take $r = \min(R)$ & so $0 \leq r = a - bm \in R$ for some $m \in \mathbb{Z}$, choose $q = m \in \mathbb{Z}$. So by construction $a = bq + r$. Note, $r < b$ as were such not true, then $r > a - b(q + 1) = r - b \in R$ but this contradicts $r = \min(R)$, so we have $0 \leq r < b$. (Uniqueness): Suppose $q, q', r, r' \in \mathbb{Z}$ with $0 \leq r', r < b$ & $a = bq + r = bq' + r'$, also WLOG take $r' \leq r$. So with $\frac{r - r'}{b} \in (-1, 1)$, & $\frac{r - r'}{b} = \frac{a - bq - (a - bq')}{b} = q' - q \in \mathbb{Z}$, we have $\frac{r - r'}{b} = 0$ i.e. $r = r'$. So $q = \frac{a - r}{b} = \frac{a - r'}{b} = q'$.

Euclidian Algorithm: Let $a, b \in \mathbb{Z}^+$, apply Th^m 1.1, to give:

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ &\vdots \end{aligned}$$

$r_{n-1} = r_nq_{n+1} + 1$. Here, $\gcd(a, b) = r_n$. To prove this relation we show:

$$G(r_i, r_{i+1}) = G(r_{i+1}, r_{i+2}),$$

where

$$G(a, b) = \{n \in \mathbb{Z} : n|a \text{ \& } n|b\}.$$

Reverse Euclid:

Given a completed algorithm we can obtain $n, m \in \mathbb{Z}$ s.t. $na + mb = \gcd(a, b)$ i.e. taking the end of a Euclidian algorithm:

$$\begin{aligned} &\vdots \\ r_{n-3} &= r_{n-2}q_{n-1} + r_{n-1} \quad (1) \\ r_{n-2} &= r_{n-1}q_n + r_n \quad (2) \\ r_{n-1} &= r_nq_{n+1} \end{aligned}$$

we have:

From (2) and (1):

$$\begin{aligned} r_n &= r_{n-2} - r_{n-1}q_n \quad (3) \\ r_{n-1} &= r_{n-3} - r_{n-2}q_{n-1} \quad (4) \end{aligned}$$

From (3) and (4):

$$\begin{aligned} r_n &= r_{n-2} - r_{n-1}q_n \\ &= r_{n-2} \\ &\quad - (r_{n-3} - r_{n-2}q_{n-1})q_n \\ &\vdots \end{aligned}$$

we continue this ladder up, simplifying each time, (Note: $r_n = \gcd(a, b)$)

Theorem 1.9. Every integer $n \geq 2$ can be written as a product of primes.

Proof. (By strong induction): (Base case, $n = 2$): Trivially true, $n = 2$ a product of primes. (Inductive Hypothesis): Suppose for $n = 2, 3, \dots, k$ $n = p_{n1}p_{n2} \dots p_{nm}n$. Then for $n = k + 1$, if $k + 1$ is prime we are done. If not, $k + 1 = ab$ where $a, b \in \{2, 3, \dots, k\}$, so by the hypothesis a and b are products of primes; hence $k + 1$ is too. So $\forall n \in \mathbb{N}$ with $n \geq 2$, we are done.

Theorem 1.10. (Euclid's property of primes) $\forall p \in \text{Primes} \ \forall a, b \in \mathbb{Z}$. $p|ab \implies p|a$ or $p|b$.

Proof. (Case $p|a$): We are done. (Case $p \nmid a$): Then using the Euclidian algorithm we obtain: $1 = na - mp$ with $n, m \in \mathbb{Z}$. So $p|b \implies p|(na - mp) \iff p|n(ab) - mbp \iff p|ab$.

the chain of implications uses a proposition in the notes which states: $p|a \ \& \ p|b \implies p|a + b \ \& \ p|ab$.

Theorem 1.12. (Fundamental Theorem of Arithmetic) $\forall n \in \mathbb{N}$ with $n \geq 2$, $\exists! p_1, p_2, \dots, p_r \in \text{Primes}$ up to ordering, s.t.: $n = p_1p_2 \dots p_r$.

Proof. The existence is from Th^m 1.9. For uniqueness, suppose $p_1p_2 \dots p_r = q_1q_2 \dots q_s$, then cancel common primes, suppose the result is not 1 i.e. we are left with $p_1p_2 \dots p_n = q_1q_2 \dots q_m$ with $p_i \neq q_j, \forall i \in \{1, \dots, n\}$ & $\forall j \in \{1, \dots, m\}$, after reindexing. Then $p_1|q_1q_2 \dots q_m$ so $\exists j \in \{1, \dots, m\}$ $p_1|q_j$, but q_j is prime so $p_1 = q_j, \hat{t}$.

Lemma 1.16. $\forall a, b \in \mathbb{R}$: $0a = a0 = 0$; $a(-b) = (-a)b = -(ab)$; & $-(-a)(-b) = ab$

Proof. $0a + 0 = 0a = (0 + 0)a = 0a + 0a$ so $0a = 0$. $ab + a(-b) = a(b - b) = a0 = 0 = (a - a)b = ab + (-a)b$ so by the Groups & Geo^m we have $-(ab) =$ the unique element s s.t. $ab - (ab) = 0$, $a(-b) = -(ab) = (-a)b$. $(-a)(-b) = -a(-b) = -(-(ab)) = ab$.

Lemma 1.18. (Subring Test) Let R be a ring & $S \subseteq R$. Then S is a subring of R iff the following hold, $\forall r, s \in S$:

- $1_R \in S$,
- $r \cdot s \in S$,
- $-r \in S$.

Proof. By 1 & 3 we have $-1_R \in S$ so $0_R = 1_R - 1_R \in S$ by 2. So $(S, +)$ is a group by the subgroup test with $(R, +)$. Commutativity of $+$, associativity of $+$ & $\times$, & distributivity of $\times$ over $+$ are all inherited from R . Note, as they share operations they share the identities of the operations, so as, $0_R \neq 1_R$ in R we have $0_S = 0_R \neq 1_R = 1_S$ in S , so (R1)-(R4) are satisfied.

Lemma 2.2. Suppose $\text{Char}(R) = n > 0$, then: (i) $n \cdot r = 0, \forall r \in R$; & (ii) if $m \in \mathbb{Z}^+$ then $m \cdot 1 = 0$ iff $n|m$.

Proof. (i): $n \cdot r = r + r + \dots + r$ n times $= r(1 + 1 + \dots + 1) = r0 = 0$

(ii): ($\implies$): Let $m \cdot 1 = 0$ with $m \in \mathbb{Z}^+$, applying the division Th^m with $m = nq + r$ where $q, r \in \mathbb{Z}$ & $0 \leq r < n$ so $0 = m \cdot 1 = (nq + r) \cdot 1 = q(n \cdot 1) + r \cdot 1 = q \cdot 0 + r \cdot 1 = r \cdot 1$, so $r = 0$ as otherwise $n > r \in \mathbb{Z}^+$ & $r \cdot 1 = 0$, a $\hat{t}$ of n 's defⁿ. ($\impliedby$): Let $n|m$ i.e. $\exists q \in \mathbb{Z}$ s.t. $m = qn$, so $m \cdot 1 = qn \cdot 1 = q(n \cdot 1) = q \cdot 0 = 0$.

Lemma 2.5. If S is a subring of R & R is a domain then S is a domain.

Proof. Let $r, s \in S$ with $rs = 0$, thus $rs = 0$ in R , hence $r = 0$ or $s = 0$.

Proposition 2.7. Suppose R is a domain. Then $R[X]$ is a domain.

Proof. We use the contrapositive of the definition of a domain [i.e.: R a domain iff $\forall 0 \neq r, s \in R, rs \neq 0$]. Let $0 \neq f, g \in R[X]$, taking the leading coefficients of f & g as $a \neq 0$ & $b \neq 0$, respectively, the leading coefficient of fg as $ab \neq 0$, hence $fg \neq 0$.

Note: This immediately (by induction on n) yields $R[X_1, \dots, X_n]$ is a domain if R is a domain.

Lemma 2.11. If R is a ring & $r \in R$ has both a left & right inverse then these equal & unique.

Proof. (Equal): Choose, $rs = 1 = tr$, then: $s = 1s = trs = t1 = t$. (Unique right inverse): Let $rs = 1 = rs'$ then $s = 1s = trs = t1 = trs' = 1s' = s'$, where t is chosen s.t. $tr = 1$. (Unique left inverse): Let $tr = 1 = t'r$ then $t = t1 = trs = 1s = t'rs = t'1 = t'$, where s is chosen s.t. $rs = 1$.

Proposition 2.13. Every division ring is a domain (& subsequently every field is an integral domain).

Proof. Let R be a division ring & suppose $r, s \in R$ & $rs = 0$. Then assume $r \neq 0$ so r^{-1} exists, then $s = r^{-1}rs = r^{-1}0 = 0$. Then given the immediate other case of $r = 0$ in all cases either $r = 0$ or $s = 0$. Hence R is a domain.

Lemma 2.12. The following conditions for an integer $n \geq 2$ are equivalent:

- $\mathbb{Z}_n$ is an integral domain;
- $\mathbb{Z}_n$ is a field;
- n is prime.

Proof. (i) $\implies$ (ii): Suppose $\mathbb{Z}_n$ is a domain & let $[m]_n \in \mathbb{Z}_n$ & $[m]_n \neq 0$. Consider the list $[0]_n[m]_n, [1]_n[m]_n, \dots, [n-1]_n[m]_n$, each element in the list is distinct as, suppose otherwise then $[i]_n[m]_n = [j]_n[m]_n$ for some distinct $i, j \in \{0, 1, \dots, n-1\}$, then $([i]_n - [j]_n)[m]_n = 0$ but $\mathbb{Z}_n$ is suppose to be a domain & $[m]_n \neq 0$ so $[i]_n - [j]_n = [0]_n$ i.e. $[i]_n = [j]_n, \hat{t}$. Then as there are n elements of the list each in $\mathbb{Z}_n$, then $\{[0]_n, [1]_n, \dots, [(n-1)]_n\} = \mathbb{Z}_n$. So some multiple of $[m]_n$ say $[r]_n \in \mathbb{Z}_n$ is s.t. $[m]_n[r]_n = [r]_n[m]_n = [1]_n$.

(ii) $\implies$ (iii): Suppose $\mathbb{Z}_n$ is a field then suppose $m, l \in \mathbb{N}$ are s.t. $ml = n$, so $0 < m \leq l \leq n$. Then $[m]_n[l]_n = [ml]_n = [n]_n = [0]_n$ so as fields are domains, $[m]_n = [0]_n = [n]_n$ or $[l]_n = [0]_n = [n]_n$ & so, by the equivalence class definition, $m = n$ or $l = n$ as neither are 0. So if $m = n, n = l$ & if $l = n, m = 1$, so

the only factors of n are 1 & n in $\mathbb{N}$, i.e. n is prime.

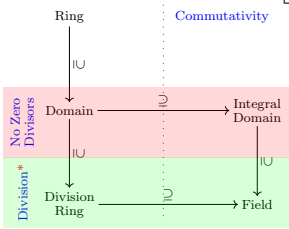
(iii) $\implies$ (i): Suppose n is prime & let $[m]_n, [l]_n \in \mathbb{Z}_n$ s.t. $[m]_n[l]_n = 0$ i.e. $[ml]_n = [0]_n$ definitionally $n | ml$ so by Th^m 1.10 $n | m$ or $n | l$ i.e. $[m]_n = [0]_n$ or $[l]_n = [0]_n$. So with $\mathbb{Z}_n$'s commutativity, $\mathbb{Z}_n$ is an integral domain.

Proposition T1. In a domain R if $ab = ac$ or $ba = ca$ for $a, b, c \in R$ & $a \neq 0$ then $b = c$.

Proof. If $ab = ac$, then $a(b - c) = 0$. If $ba = ca$, then $(b - c)a = 0$. Since $a \neq 0$ and R is a domain, $b - c = 0$.

Lemma 2.15. In a ring R , the set of units R^* forms a group under multiplication.

Proof. (G0): $1 \cdot 1 = 1$ so $1 \in R^*$ i.e. $R^* \neq \emptyset$; (G1): given (R2) we have associativity & given $r \in R^*$, $(rs)(s^{-1}r^{-1}) = r(ss^{-1})r^{-1} = 1$ & $(s^{-1}r^{-1})(rs) = s^{-1}(r^{-1}r)s = 1$, so $rs \in R^*$ as required; (G2): (R4); & (G3): Let $r \in R^*$ so r is a unit so $r^{-1} \in R^*$ exists, which itself has inverse $(r^{-1})^{-1} = r \in R$, so $r^{-1} \in R^*$, as required.



Lemma 3.3. Suppose $\phi: R \rightarrow S$ is an isomorphism. Then: (i) $\phi(1_R) = 1_S$; (ii) $\phi(0_R) = 0_S$; (iii) $\phi(-r) = -\phi(r), \forall r \in R$; (iv) $r \in R$ is invertible iff $\phi(r) \in S$ is invertible and, in that case, $(\phi(r))^{-1} = \phi(r^{-1})$; (v) $r \in R$ is nilpotent iff $\phi(r)$ is nilpotent and, in that case, they have the same index of nilpotence.

Proof. (i): Let $s \in S$. Since ϕ is bijective, $s = \phi(r)$ for some $r \in R$. Then $\phi(1_R)s = \phi(1_R)\phi(r) = \phi(r) = s$ and similarly $s\phi(1_R) = s$, hence $\phi(1_R) = 1_S$. (ii): $\phi(0_R) + 1_S = \phi(0_R + 1_R) = \phi(1_R) = 1_S$, so $\phi(0_R) = 0_S$. (iii): $\phi(-r) + \phi(r) = \phi(0_R) = 0_S$, so $\phi(-r) = -\phi(r)$. (iv): If r is invertible then $\phi(r^{-1})\phi(r) = \phi(r^{-1}r) = 1_S$ and $\phi(r)\phi(r^{-1}) = \phi(rr^{-1}) = 1_S$, so $\phi(r)$ is invertible with $(\phi(r))^{-1} = \phi(r^{-1})$. Conversely, apply this to $\phi^{-1}: S \rightarrow R$. (v): Since $(\phi(r))^n = \phi(r^n)$, $r^n = 0_R$ iff $(\phi(r))^n = 0_S$. Thus the least such n is the same.

Lemma 3.7. Suppose $\theta: R \rightarrow S$ is a homomorphism. Then: (i) $\theta(0) = 0$; (ii) $\theta(-r) = -\theta(r), \forall r \in R$; (iii) if $r \in R$ is invertible, then $\theta(r) \in S$ is invertible and $(\theta(r))^{-1} = \theta(r^{-1})$; (iv) if $r \in R$ is nilpotent, then $\theta(r)$ is nilpotent with index $\leq$ the index of r ; (v) $\text{Im}(\theta)$ is a subring of S .

Proof. (i): As in Lem^m 3.3(ii). (ii): As in Lem^m 3.3(iii). (iii): If r is invertible then $\theta(r^{-1})\theta(r) = \theta(r^{-1}r) = 1_S$ and similarly $\theta(r)\theta(r^{-1}) = 1_S$, so $(\theta(r))^{-1} = \theta(r^{-1})$. (iv): If r is nilpotent of index n , then $\theta(r)^n = \theta(r^n) = \theta(0) = 0$, so $\theta(r)$ has index at most n . (v): Since $\theta(1_R) = 1_S, 1_S \in \text{Im}(\theta)$. Let $s = \theta(r), s' = \theta(r') \in \text{Im}(\theta)$. Then $s + s' = \theta(r + r'), ss' = \theta(rr')$, and $-s = -\theta(r) = \theta(-r)$, so by Lem^m 1.18, $\text{Im}(\theta)$ is a subring of S .

Lemma 3.10. (i) If $\theta: R \rightarrow S$ & $\beta: S \rightarrow T$ are homomorphisms then $\beta \circ \theta: R \rightarrow T$ is a homomorphism. (ii) If θ, β are embeddings then $\beta \circ \theta$ is an embedding. (iii) If θ, β are homomorphisms & $\beta \circ \theta$ is an embedding then θ is an embedding.

Proof. (i): Let $r, r' \in R$. Then $\beta \circ \theta(r + r') = \beta(\theta(r + r')) = \beta(\theta(r) + \theta(r')) = \beta \circ \theta(r) + \beta \circ \theta(r')$

Similarly, $\beta \circ \theta(rr') = \beta \circ \theta(r)\beta \circ \theta(r')$, and $\beta \circ \theta(1_R) = \beta(\theta(1_R)) = \beta(1_S) = 1_T$ hence $\beta \circ \theta$ is a homomorphism. (ii) & (iii): follow from properties of injective maps.

Lemma 3.15. If $\theta: R \rightarrow S$ is a homomorphism then θ is injective iff $\ker(\theta) = 0$.

Proof. ($\implies$) Suppose θ is injective. Then $\theta(0) = 0$ by Lem^m 3.7(i), then

let $r \in \ker(\theta)$ then $\theta(r) = 0 = \theta(0)$ so $r = 0 \in \{0\}$, thus $\ker(\theta) = \{0\}$. ($\impliedby$) Suppose $\ker(\theta) = 0$. Now let $r, r' \in R$ s.t. $\theta(r) = \theta(r')$ thus $\theta(r - r') = \theta(r) - \theta(r') = 0$ i.e. $r - r' \in \ker(\theta) = \{0\}$ so $r - r' = 0$, thus $r = r'$, as required.

Lemma 3.17. (i) Suppose $\theta: R \rightarrow S$ is a homomorphism. Then $\ker(\theta)$ is a subgroup of $(R, +)$. (ii) Let $r, r' \in R$. Then $\theta(r) = \theta(r')$ iff $r - r' \in \ker(\theta)$ iff r, r' belong to the same coset of $\ker(\theta)$ in R .

Proof. (i): $0 \in \ker(\theta)$ so it is non-empty. Let $r, r' \in \ker(\theta)$. Then $\theta(r + r') = \theta(r) + \theta(r') = 0 + 0 = 0$, so $r + r' \in \ker(\theta)$. Also $\theta(-r) = -\theta(r) = 0$, so $-r \in \ker(\theta)$. Hence by the subgroup test, $\ker(\theta) \leq (R, +)$. (ii): Let $r, r' \in R$. Then $\theta(r) = \theta(r') \iff \theta(r - r') = 0 \iff r - r' \in \ker(\theta) \iff r + \ker(\theta) = r' + \ker(\theta)$.

Proposition 3.22. A commutative ring R is a field iff the only ideals of R are R and $\{0\}$.

Proof. ($\implies$) Suppose R is a field, then clearly $\{0\} \triangleleft R$ now suppose $\{0\} \neq I \triangleleft R$ so then by definition I contains a non-zero element of R say $r \in I$. Now let $s \in R$, so, as R is a field r^{-1} exists, then $s = (sr^{-1})r \in I$, hence $I = R$, as required. ($\impliedby$) Suppose the only ideals of R are R and $\{0\}$. Now let $r \in R \setminus \{0\}$ then $\langle r \rangle = R$ hence $1 \in \langle r \rangle$ so by the definition of the principle ideal in a commutative ring $1 = sr (= rs)$ for some $s \in R$, hence s is r 's inverse all non-zero elements in R have inverses, hence R is a field.

Proposition 3.24. If $\theta: R \rightarrow S$ is a homomorphism of rings, then $\ker(\theta)$ is an ideal of R .

Proof. We show $\ker(\theta)$ satisfies the three conditions for an ideal. Firstly $0 \in \ker(\theta)$. Let $r, r' \in \ker(\theta)$. Then $\theta(r + r') = \theta(r) + \theta(r') = 0 + 0 = 0$, so $r + r' \in \ker(\theta)$. Let $s \in R$. Then $\theta(sr) = \theta(s)\theta(r) = \theta(s)0 = 0$ and similarly $\theta(rs) = 0$. Therefore $sr, rs \in \ker(\theta)$.

Corollary 3.25. If $\theta: R \rightarrow S$ is a homomorphism of rings & R is a field then θ is a monomorphism.

Proof. By ProP 3.24, $\ker(\theta) \triangleleft R$, and so ProP 3.22 implies $\ker(\theta) = \{0\}$ or R . We cannot have $\ker(\theta) = R$, as then $1 = \theta(1) = 0$. Therefore $\ker(\theta) = \{0\}$ and θ is a monomorphism.

Proposition 3.26. Suppose I and J are ideals of the ring R . Then: (i) $I + J = \{a + b : a \in I, b \in J\}$ is an ideal; (ii) $I \cap J$ is an ideal; (iii) if $\{I_\lambda\}_\lambda$ is any collection of ideals of R , then $\bigcap_\lambda I_\lambda$ is an ideal.

Proof. (i): $0 \in I + J$, since $0 \in I$ and $0 \in J$. Let $c, c' \in I + J$, say $c = a + b, c' = a' + b'$, for $a, a' \in I, b, b' \in J$. Then $c + c' = (a + b) + (a' + b') = (a + a') + (b + b') \in I + J$. If $r \in R$, then $rc = r(a + b) = ra + rb \in I + J$ and similarly $cr \in I + J$. Therefore $I + J \triangleleft R$.

(ii): Let $K = \bigcap_\lambda I_\lambda$. Since $0 \in I_\lambda$ for every $\lambda, 0 \in K$. Let $a, b \in K$. Then $a, b \in I_\lambda$ for every λ , so $a + b \in I_\lambda$ for every λ , hence $a + b \in K$. If $r \in R$, then $ra, ar \in I_\lambda$ for every λ , hence $ra, ar \in K$. Thus $K \triangleleft R$. (iii): This is the special case of (iii) with the collection $\{I, J\}$.

Lemma 4.2. (i) The operations $+$ and $\times$ on R/I are well-defined, i.e. if $r + I = r' + I$ and $s + I = s' + I$, then $(r + s) + I = (r' + s') + I$ and $rs + I = r's' + I$. (ii) With these operations, R/I forms a ring. The zero element is $0 + I (= I)$, the identity is $1 + I$, and $-(r + I) = (-r) + I$. If r^{-1} exists in R , then $r + I$ is invertible in R/I with inverse $r^{-1} + I$.

Proof. (i): Let $r + I = r' + I$ and $s + I = s' + I$, so $r - r', s - s' \in I$. Then $(r - r') + (s - s') = (r + s) - (r' + s') \in I$, hence $(r + s) + I = (r' + s') + I$. Also $(r - r')s' \in I$ and $r'(s - s') \in I$, so $rs - r's' = (r - r')s' + r'(s - s') \in I$, hence $rs + I = r's' + I$.

(ii): By definition of $+$ and $\times$, $(r + I) + (0 + I) = r + I$ and $(r + I)(1 + I) = (1 + I)(r + I) = r + I$. Associativity, distributivity, and commutativity of $+$ are induced from R . We have $(r + I) + ((-r) + I) = 0 + I$, so $(R/I, +)$ is an abelian group. Finally, if r^{-1} exists in R , then $(r + I)(r^{-1} + I) = 1 + I = (r^{-1} + I)(r + I)$.

Lemma 4.4. Let I be a proper ideal of R . (i) The map $\pi: R \rightarrow R/I, \pi(r) = r + I$, is a surjective ring homomorphism with kernel I . (ii) Let $\theta: R \rightarrow S$ be a homomorphism with $I \subseteq \ker(\theta)$. There is an induced homomorphism $\theta': R/I \rightarrow S$ with $\theta' \circ \pi = \theta$.

Proof. (i): Every element of R/I has form $r + I = \pi(r)$, so π is surjective. Also

$\pi(r + s) = (r + s) + I = (r + I) + (s + I) = \pi(r) + \pi(s)$, similarly $\pi(rs) = \pi(r)\pi(s)$, and π

Theorem 5.1. (Division Theorem for Polynomials) Let K be a field and $f, g \in K[X]$ with $g \neq 0$. Then there are unique $q, r \in K[X]$ with $f = qg + r$ and $\deg(r) < \deg(g)$ or $r = 0$.

Proof. Let $A = \{f - hg : h \in K[X]\}$. Choose $r = f - qg \in A$ of least degree. If $r = 0$ or $\deg(r) < \deg(g)$, done. Otherwise, since K is a field, a suitable scalar multiple of $\chi \deg r - \deg g$ can be subtracted from r to cancel its leading term, giving $r - hg = f - (q + h)g \in A$ of smaller degree, contradiction. For uniqueness, if $qg + r = q'g + r'$, then $(q - q')g = r' - r$. If $q \neq q'$, the left has degree at least $\deg(g)$, while the right has degree $< \deg(g)$, impossible. Thus $q = q'$ and $r = r'$. $\square$

Corollary 5.4. Let K be a field, $f \in K[X]$, and $a \in K$. Then a is a root of f iff $X - a$ is a factor of f .

Proof. If $X - a \mid f$, then $f(a) = 0$. Conversely, suppose $f(a) = 0$. Divide f by $X - a$: $f = q(X - a) + r$, where $r = 0$ or $\deg r < 1$, so r is constant. Substituting $X = a$ gives $0 = f(a) = r$. Hence $f = q(X - a)$. $\square$

Corollary 5.6. Let K be a field and $f, g \in K[X]$. Then the ideal generated by f and g equals the ideal generated by their greatest common divisor, i.e. $\langle f, g \rangle = \langle \gcd(f, g) \rangle$.

Theorem 5.8. Let K be a field. Then every ideal of $K[X]$ is principal.

Proof. Let $I \triangleleft K[X]$. If $I = \{0\} = (0)$, then I is principal. Otherwise choose non-zero $g \in I$ of least degree. Let $f \in I$. By Theorem 5.1, $f = qg + r$ with $r = 0$ or $\deg(r) < \deg(g)$. Since $r = f - qg \in I$, minimality of $\deg(g)$ forces $r = 0$. Hence every $f \in I$ is a multiple of g , so $I = (g)$. $\square$

Lemma 6.2. Let K be a field and $f \in K[X]$ be irreducible. Then $\langle f \rangle$ is a maximal ideal of $K[X]$.

Proof. Let $\langle f \rangle \leq I \leq K[X]$. By Th^m 5.8, $I = \langle g \rangle$. Since $f \in \langle g \rangle$, $f = qg$ for some $q \in K[X]$. As f is irreducible, q or g is invertible. If q is invertible, then $\langle g \rangle = \langle f \rangle$. If g is invertible, then $\langle g \rangle = K[X]$. Hence $\langle f \rangle$ is maximal. $\square$

Theorem 6.3. (Kronecker's Theorem) Let K be a field and $f \in K[X]$ be irreducible of degree n . Define $L = K[X]/\langle f \rangle$. Then: (i) L is a field and $\pi : K[X] \rightarrow L$ induces an embedding $\theta : K \rightarrow L$; (ii) $\alpha = X + \langle f \rangle$ is a root of f in L ; (iii) L is a vector space over K of dimension n , with basis $\{1, \alpha, \dots, \alpha^{n-1}\}$.

Proof. (i): By Lemma 6.2, $\langle f \rangle$ is maximal, so L is a field by Corollary 4.13. Define $\theta : K \rightarrow L$ by $\theta(a) = a + \langle f \rangle$. This is a homomorphism. If $\theta(a) = \theta(b)$, then $a - b \in \langle f \rangle$. Since $\deg(f) \geq 1$, this forces $a - b = 0$, so θ is injective. (ii): Let $f = b_0 + b_1X + \dots + b_nX^n$. For $\alpha = X + \langle f \rangle$, $f(\alpha) = b_0 + b_1\alpha + \dots + b_n\alpha^n = f + \langle f \rangle = (f) = 0_L$. Thus α is a root of f in L . (iii): If $g + \langle f \rangle \in L$, divide g by f : $g = qf + r$, with $r = 0$ or $\deg(r) < n$. Writing $r = a_0 + a_1X + \dots + a_{n-1}X^{n-1}$, we get $g + \langle f \rangle = r + \langle f \rangle = a_0 + a_1\alpha + \dots + a_{n-1}\alpha^{n-1}$. Thus L is spanned by $1, \alpha, \dots, \alpha^{n-1}$. Linear independence follows since any non-zero polynomial of degree $< n$ cannot lie in $\langle f \rangle$. $\square$

Definitions

Definition 1.4. $\forall a, b \in \mathbb{Z}$, $\gcd(a, b) = \max\{n \in \mathbb{Z} : n|a \ \& \ n|b\}$, noting $\gcd(0, 0) = 0$ & if $\gcd(a, b) = 1$ then a & b are coprime.

Definition 1.6. $\forall a, b, n \in \mathbb{Z}$ with $n \geq 2$. a is **congruent** to b **modulo** n iff $n|a - b$, written $a \equiv b \bmod n$. Then define the congruence class of a w.r.t. n as: $[a]_n = \{b \in \mathbb{Z} : b \equiv a \bmod n\} = \{b \in \mathbb{Z} : n \mid (b - a)\} = \{a + kn : k \in \mathbb{Z}\}$

We thus have: $\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\} = \{[0]_n, [1]_n, \dots, [n-1]_n\}$ Note: $\tilde{b} \equiv a \bmod n$ iff $\tilde{b} = a + kn$ iff $[a]_n = [a + nk : k \in \mathbb{Z}]$ iff $[a]_n = [a + kn]_n$

Definition 1.14. A **ring** is a set R with two binary operations $+$ & $\times$, defined on R , satisfying the following:

- (R1) $(R, +)$ is an abelian group with identity 0.
- (R2) $\times$ is associative.
- (R3) $\times$ distributes over $+$ i.e.: $a(b + c) = ab + ac$, $(a + b)c = ac + bc$, $\forall a, b, c \in R$
- (R4) The multiplicative identity 1 $\in R$ & $1 \neq 0$

Example. With $n \geq 2$, $M_n(R)$ is the ring of $n \times n$ matrices, a non-commutative ring. Note: the only two-sided ideals of $M_n(F)$, F a field, are $M_n(F)$ & $\{0\}$ but this isn't a division ring hence even relaxing commutative in Prop³ 3.22 doesn't yield the same prop³ with field $\rightarrow$ division ring. **Example.** For X a set $\mathcal{P}(X)$ with $A + B = (A \cup B) \setminus (A \cap B)$,

$AB = A \cap B$ & $0 = \emptyset, -A = A, A, B \in \mathcal{P}(X)$. **Definition 1.17.** Let R be a ring & $S \subseteq R$. Then S is a **subring** of R if S is a ring w.r.t. the same addition & multiplication as R & contains the multiplicative identity of R , 1_R . **Definition 1.21.** Let R be a ring. The **ring of polynomials** $R[X] = \{\sum_{i \geq 0} a_i X^i : \text{a finite sum : } a_i \in R\}$ with addition & multiplication defined as: $\sum_{i \geq 0} a_i X^i + \sum_{i \geq 0}^m b_i X^i = \sum_{i \geq 0}^{\max(n,m)} (a_i + b_i) X^i$ & $(\sum_{i \geq 0}^n a_i X^i)(\sum_{i \geq 0}^m b_i X^i) = n+m \sum_{j=0}^{} (\sum_{i+j=j} a_j b_{i-j}) X^i$ (taking missing coefficients as 0) $0 = \sum_{i \geq 0} 0X^i$ &

$1 = 1X^0 + \sum_{i \geq 1} 0X^i$. Then the **degree** of a polynomial is the greatest $i \in \mathbb{N} \cup \{0\}$ s.t. $a_i \neq 0$ or if no such i exist $\deg(0) = -\infty$ More generally, we have: $R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$ $\forall f, g \in R[X]$, $\deg(f + g) \leq \max(\deg(f), \deg(g))$ & $\deg(fg) \leq \deg(f) + \deg(g)$. **Definition 1.24.** If R_1 & R_2 are rings then the **Cartesian product** $R_1 \times R_2$ with the $+$ & $\times$ defined element-wise is a ring.

Definition 2.1. For $a \in R$ define: for $n \in \mathbb{Z}^+$, $n \cdot a = a + a + \dots + a$;

n times $0 = \sum_{i \geq 0} 0X^i$ & for $n \in \mathbb{Z}^-$, $n \cdot a = (-n) \cdot (-a)$. The **Characteristic**, $\text{char}(R)$, of a ring R is the least positive integer n s.t. $n \cdot 1 = 0$, or if no exist $\text{char}(R) = 0$. Note: $(n + m) \cdot a = n \cdot a + m \cdot a$, $n \cdot (a + b) = n \cdot a + n \cdot b$, & $n \cdot (m \cdot a) = (nm) \cdot a$, all easily verifiable.

Definition 2.3. A non-zero element $r \in R$ is a **zero-divisor** if $\exists s \in R$ a non-zero element s.t. $rs = 0$ or $sr = 0$.

A ring R is a domain if, $\forall r, s \in R$: $rs = 0 \implies r = 0$ or $s = 0$. i.e. a ring with no zero divisors is a domain & vice versa.

Definition 2.9. A **division ring** is a ring where every non-zero element has a right & left inverse, i.e.: $\forall r \in R \setminus \{0\}$, $\exists s, t \in R$ s.t. $rs = 1 = tr$ with s, t being the right & left inverses, respectively. Note: this immediately collapse, due to Le^m 2.11, to just there exists an inverse as $s = t$. An **invertible** element r is called a **unit**.

Example. Division rings are not all fields e.g. the quaternions $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ where $i^2 = j^2 = k^2 = -1$ & $ij = k$, $jk = i$, & $ki = j$, this is non-commutative as $-ij = -k = jjk = ji$ so $ij \neq ji$ and this is a division ring as for $z = a + bi + cj + dk$ $z^{-1} = \frac{a-bi-cj-dk}{a^2+b^2+c^2+d^2}$, taking $z \neq 0$.

Definition 2.16. An element r of a ring R is **nilpotent** if $\exists n \in \mathbb{Z}^+$ s.t. $r^n = 0$, and the least such n is the **index of nilpotence** of r .

An element $r \in R$ is **idempotent** if $r^2 = r$.

Note: In any ring R , $0R$ is nilpotent with index 1 and $0R$ and 1_R are idempotent.

Definition 3.1. If R & S are rings then an **isomorphism** from R to S is a bijection $\phi : R \rightarrow S$ s.t.

$\forall r, r' \in R$: $\phi(r + r') = \phi(r) + \phi(r')$,

$\phi(rr') = \phi(r)\phi(r')$

$\exists \phi : R \rightarrow S$ an isomorphism iff R & S are **isomorphic**, written $R \simeq S$. **Example.** For 4 element rings $\mathbb{Z}_4 \not\cong \mathbb{Z}_2 \times \mathbb{Z}_2$, since given $[2]_4 \neq 0$ and $[2]_4^2 = [0]_4$, $[2]_4$ is nilpotent, where as there are no non-zero nilpotent elements in $\mathbb{Z}_2 \times \mathbb{Z}_2$, hence the result by Le^m 3.3.

Definition 3.6. If R & S are rings then a **homomorphism** from R to S is a map $\vartheta : R \rightarrow S$ s.t.

$\forall r, r' \in R$:

$\vartheta(r + r') = \vartheta(r) + \vartheta(r')$,

$\vartheta(rr') = \vartheta(r)\vartheta(r')$,

$\vartheta(1_R) = 1_S$.

Example. The map $\mathbb{Z} \rightarrow \mathbb{Z}_n$ by $m \mapsto [m]_n$ is a homomorphism. Also, $\vartheta : \mathbb{Z}_{kn} \rightarrow \mathbb{Z}_n$ by $\vartheta([m]_{kn}) = [m]_n$ is a homomorphism, well definedness must be shown, and is true as; let

$[m]_{kn} = [m']_{kn} \in \mathbb{Z}_{kn}$ then $kn \mid m - m'$ so $n \mid m - m'$ thus $[m]_n = [m']_n$.

Definition 3.9. An **embedding** or **monomorphism** is an injective homomorphism. There is a trivial embedding along each arrow of $\mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{C}[X]$.

Example. With R a ring, the map $R[X] \rightarrow R[X]$ by $p(X) \mapsto p(q(X))$ for some $q(X) \in R[X]$.

Definition 3.12. If $\vartheta : R \rightarrow S$ is a ring homomorphism, then the **kernel** of ϑ is defined as $\ker(\vartheta) = \{r \in R : \vartheta(r) = 0_S\}$.

Definition 3.16. An **automorphism** of a ring is an isomorphism from and to itself.

Example. $\mathbb{Z}[i] \rightarrow \mathbb{Z}[i]$ by

$a + bi \mapsto a - bi$ is an automorphism.

Definition 3.18. An **ideal** of a ring R is a subset $I \subseteq R$ s.t.: $0 \in I$; $a + b \in I$, $\forall a, b \in I$; & $ar, ra \in I$, $\forall a \in I$, $\forall r \in R$. We write $I \triangleleft R$ to mean I is an ideal of R .

Definition 3.19. If $a \in R$ then $\{r_1as_1 + \dots + r_nas_n : n \geq 1, r_i, s_i \in R\}$ is an ideal containing a and is the smallest ideal of R containing a . It is called the **principal ideal generated by** a , denoted $\langle a \rangle$. If R is commutative then $\langle a \rangle = \{ar : r \in R\}$. A **principal ideal** is one which can be generated by a single element.

Examples 3.20. (i) In every ring $\langle 0 \rangle = \{0\}$ is the smallest ideal, the **trivial ideal**. (ii) In every ring $\langle 1 \rangle = R$ is an ideal and every other ideal $\neq R$ is a **proper ideal**. (iii) If $n \in \mathbb{Z}$ then $\langle n \rangle = \{nk : k \in \mathbb{Z}\}$. (iv) If $p \in R[X]$ then $\langle p \rangle$ is the set of polynomials with p as a factor.

Definition 3.21. A **right ideal** is defined as for a general ideal but with the third condition replaced by the weaker condition: $a \in I$ and $r \in R$ implies $ar \in I$. A **left ideal** is defined similarly. If $a \in R$, the **principal right ideal generated by** a is $\{ar : r \in R\}$, denoted aR . **Example.** Let $R = M_2(\mathbb{Z})$ and

$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ then by simple

multiplication the principle left and right ideals are $\{\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} : a, b \in \mathbb{Z}\}$

and $\{\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{Z}\}$ whilst $\langle A \rangle = M_2(\mathbb{Z})$ as

$I_2 = A + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} A \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \langle A \rangle$

Note: If $1 \in I$ with $I \triangleleft R$ the $\forall r \in R$ $r = r1 \in I$, so $I = R$.

Definition 3.22. Let $A \subseteq R$ and R be a commutative ring, then $\{a_1r_1 + \dots + anr_n : n \geq 1, a_i \in A, r_i \in R\}$ is the **ideal generated by** A .

Example. Working in $\mathbb{Z}[X]$ the ideal $I = \langle 2, X \rangle := \langle \{2, X\} \rangle$ is not a principal ideal. Suppose $I = \langle p \rangle$. Since $2 \in I$, $2 = pq$ for some $q \in \mathbb{Z}[X]$. As $\mathbb{Z}$ is a domain, $0 = \deg(2) = \deg(pq) = \deg(p) + \deg(q)$, so $p, q \in \mathbb{Z}$ and $p = \pm 1$ or $p = \pm 2$. If $p = \pm 1$, then $1 = 2r_1 + Xr_2$ for some $r_1, r_2 \in \mathbb{Z}[X]$, impossible since the RHS has even constant term. Hence $p = \pm 2$. But $X \in I$, so $X = 2r_3$ for some $r_3 \in \mathbb{Z}[X]$, impossible since $2r_3$ has even coefficients. Thus I is not principal. **Definition 3.26.** We can also define a **product of ideals**: if $I, J \triangleleft R$, then their product is the ideal generated by $\{ab : a \in I, b \in J\}$. This set will not in general be closed under addition, hence we say "generated by". Powers of an ideal are defined inductively by $I^n + 1 = I^n I$.

Definition 2.16. Let R be a ring and let I be a proper ideal. Let R/I denote the set of cosets of I in the additive group $(R, +)$:

$R/I = \{r + I : r \in R\}$. Define operations $+$ and $\times$ on R/I by

$(r + I) + (s + I) = (r + s) + I$, $(r + I)(s + I) = rs + I$.

Definition 4.1. The ring R/I is called the **factor ring** (or **quotient ring**) of R by I .

Example. $\mathbb{Z}/(n) = \mathbb{Z}_n$.

Exercise. Write out the elements of the factor ring $\mathbb{Z}_6/\langle [3]_6 \rangle$. What is this factor ring isomorphic to?

Definition 4.4. The map $\pi : R \rightarrow R/I$, $\pi(r) = r + I$, is called the **canonical projection**, or **canonical surjection**.

Example. 4.6. Take $R = \mathbb{Z}$ and $I = \langle 6 \rangle = 6\mathbb{Z}$, so the canonical projection is $\pi : \mathbb{Z} \rightarrow \mathbb{Z}_6$. Let $\vartheta : \mathbb{Z} \rightarrow \mathbb{Z}/\langle 2 \rangle = \mathbb{Z}_2$ be the canonical projection. Then $\ker(\vartheta) = \langle 2 \rangle \supseteq \langle 6 \rangle$, so by Le^m 4.4 there is a unique homomorphism θ' with $\theta' \circ \pi = \vartheta$. This is the map $\theta' : \mathbb{Z}_6 \rightarrow \mathbb{Z}_2$ already seen in Examples 3.8.

Example. 4.7. Let $R = \mathbb{R}[X]$ and $I = \langle X^2 + 1 \rangle$. Let $\pi : \mathbb{R}[X] \rightarrow \mathbb{R}[X]/\langle X^2 + 1 \rangle$ be the canonical surjection. Let $\vartheta : \mathbb{R}[X] \rightarrow \mathbb{C}$ be evaluation at i , $\vartheta(p(X)) = p(i)$. Then $p \in \ker(\vartheta)$ iff $X^2 + 1 \mid p$, so $\ker(\vartheta) = \langle X^2 + 1 \rangle$. By Le^m 4.4, there is a unique map $\theta' : \mathbb{R}[X]/\langle X^2 + 1 \rangle \rightarrow \mathbb{C}$ through

which ϑ factors. It is injective and surjective, hence an isomorphism.

Thus $\mathbb{R}[X]/\langle X^2 + 1 \rangle \simeq \mathbb{C}$.

Notation: if $I \subseteq J$ are ideals then usually write $I \leq J$ ($J \geq I$) instead of subset notation, with the same meaning.

Example. 4.9. The ideals of $\mathbb{Z}_6 = \mathbb{Z}/\langle 6 \rangle$ are in one-to-one correspondence with the ideals of $\mathbb{Z}$ containing $\langle 6 \rangle$. There are 4 of each: $\langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 6 \rangle$ correspond to $\pi(\mathbb{Z}) = \mathbb{Z}_6 = \{[0]_6, [1]_6, [2]_6, [3]_6, [4]_6, [5]_6\}$, $\pi(\langle 2 \rangle) = \langle [2]_6 \rangle = \{[0]_6, [2]_6, [4]_6\}$, $\pi(\langle 3 \rangle) = \langle [3]_6 \rangle = \{[0]_6, [3]_6\}$, $\pi(\langle 6 \rangle) = \langle [0]_6 \rangle = \{[0]_6\}$.

The notation J/I is also used instead of πJ for the image of J in R/I .

Example. 4.11. Since $\langle 6 \rangle \leq \langle 2 \rangle$, Th^m 4.10 gives $(\mathbb{Z}/\langle 6 \rangle)/(\langle 2 \rangle/\langle 6 \rangle) \simeq \mathbb{Z}/\langle 2 \rangle$.

Definition 4.12. An ideal I of a ring R is **maximal** if it is proper and for any ideal J with $I \leq J \leq R$, either $J = I$ or $J = R$.

Definition 4.14. A proper ideal I of a commutative ring R is **prime** if whenever $r, s \in R$ and $rs \in I$, then $r \in I$ or $s \in I$.

Example. 4.15. All ideals of $\mathbb{Z}$ are principal. The prime ideals are precisely the ideals generated by a prime number or zero.

Example. 5.2. In $\mathbb{Z}_5[X]$,

$X^4 + 2X^2 + X + 4 = (X^2 + 2X + 3)(X^2 + 3X + 3) + X$.

Definition 5.3. An element $a \in K$ is a **root** (or **zero**) of $f \in K[X]$ iff $f(a) = 0$.

Definition 5.5. The **greatest common divisor** of polynomials f, g is a polynomial d such that $d \mid f$ and $d \mid g$, and if $h \mid f$ and $h \mid g$, then $h \mid d$. This determines d up to a non-zero scalar; for uniqueness take d **monic**.

Division Algorithm for Polynomials. For $f, g \in K[X]$,

$g \neq 0$, repeatedly apply Th^m 5.1: $f = q_1g + r_1, \quad \deg(r_1) < \deg(g),$ $g = q_2r_1 + r_2, \quad \deg(r_2) < \deg(r_1),$ $r_1 = q_3r_2 + r_3, \quad \deg(r_3) < \deg(r_2),$

$\vdots$

$r_{n-1} = q_{n+1}r_n$. Since degrees strictly decrease, some $r_n + 1 = 0$. Then

$\gcd(f, g) = \gcd(g, r_1) = \dots = \gcd(r_{n-1}, r_n) = r_n$ is up to multiplication by a non-zero scalar; take the monic version.

Tracking back expresses $\gcd(f, g) = kf + lg$ for some $k, l \in K[X]$.

Factorisation. Here R is commutative.

Definition 5.9. (i) $r \in R$ is **irreducible** if r is not invertible and whenever $r = st$, either s or t is invertible. (ii) $r, s \in R$ are **associated** if $s = ur$ for some invertible $u \in R$.

Examples 5.10. (i) In $\mathbb{Z}$, irreducibles are $\pm p$ for p prime; r, s are associated iff $r = \pm s$. (ii) In $R[X]$, irreducibles are non-constant polynomials with no factor g satisfying $1 \leq \deg(g) < \deg(f)$.

Definition 5.11. A commutative domain R is a **Unique**

Factorisation Domain (UFD) if every non-zero, non-invertible element factors uniquely into irreducibles, up to rearrangement and associated factors.

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